

Property Taxonomy: When Do Conformer Statistics Help?

Category	Datasets	Finding	Verdict
Solvation Properties	ESOL FreeSolv	Hybrid improves RMSE by 4-10%	
Steric / Boltzmann	Kraken	Attention > DKO > Mean <i>(partial benefit)</i>	
Electronic Properties	QM9-Gap BDE Drugs-75K	No improvement from conformer statistics	

Conformer ensemble statistics are most useful for properties that depend on molecular shape in solution.